Worksheet 8A: Conditional Probability and Bayes' Theorem — Solutions

Series 8: Probability Deepened
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Instructor Copy — Not for Distribution

Note to instructor: Key answers are given below. Accept any mathematically equivalent form unless the question specifies otherwise.

Part A

1. (a) $4/10 = 2/5$ (b) $3/9 = 1/3$ (c) $(2/5)(1/3) = 2/15$
2. (a) $18/30 = 3/5$ (b) $6/18 = 1/3$ (c) $6/12 = 1/2$
3. (a) $P(A) = 5/36$ (b) $P(A \mid B) = P(\{(4,4)\})/P(B) = (1/36)/(1/6) = 1/6$ (c) Yes; knowing the first die is 4 makes the second die the only unknown.
4. (a) $150/260 \approx 0.577$ (b) $60/150 = 0.4$ (c) $60/120 = 0.5$ (d) $30/140 \approx 0.214$ (e) $P(\mathrm{F} \mid \mathrm{E}) = 0.4 \neq P(\mathrm{F}) = 120/260 \approx 0.462$ — not independent.
5. $P(A) = 4/52 = 1/13$; $P(B) = 12/52 = 3/13$; $P(A \cap B) = 4/52$ (all Kings are face cards); $P(A \mid B) = 4/12 = 1/3$; $P(B \mid A) = 1$ (every King is a face card).
6. $P(D) = 0.05$; $P(D^c) = 0.95$; $P(+ \mid D) = 0.90$; $P(+ \mid D^c) = 0.02$; $P(D \cap +) = 0.045$; $P(D^c \cap +) = 0.019$.
7. From table: $P(\neg E \mid F) = (40 + 20)/120 = 60/120 = 0.5$. Complement: $1 - P(E \mid F) = 1 - 0.5 = 0.5$. ✓
8. $P(3 \text{ Aces}) = (4/52)(3/51)(2/50) = 24/132600 \approx 0.000181$
9. (a) $P(A \mid B) = 0.2/0.5 = 0.4$; $P(B \mid A) = 0.2/0.4 = 0.5$ (b) $P(A \mid B) = 0.4 = P(A)$ — yes, independent.
10. $P(A \cap B) = 0.6 \times 0.3 = 0.18$

Part B

11. (a) $P(A) = 1/2$, $P(B) = 1/2$, $P(A \cap B) = 1/4$ (b) $1/4 = (1/2)(1/2)$. ✓ Independent.
12. $P(F \cap D) = 40/260$; $P(F) \cdot P(D) = (120/260)(60/260) \approx 0.1065$; $40/260 \approx 0.1538$. Not equal — not independent.
13. (a) $(0.99)^5 \approx 0.951$ (b) $1-(0.99)^5 \approx 0.049$ (c) $(0.99)^2(0.01)(0.99)^0 = (0.9801)(0.01) \approx 0.0098$
14. $(0.5)^3 = 0.125$

15. $P(C) = P(A \cap B) + P(A^c \cap B^c) = 1/4 + 1/4 = 1/2$. $P(A)P(C) = (1/2)(1/2) = 1/4 = P(A \cap C)$ ✓. But $P(A \cap B \cap C) = P(A \cap B) = 1/4$ while $P(A)P(B)P(C) = (1/2)^3 = 1/8 \neq 1/4$.
16. $0.8 \times 0.7 \times 0.6 \times 0.3 = 0.1008$
17. $0.05 \times 0.2 \times 0.3 \times 0.7 = 0.0021$
18. $P(\text{spam}) = 0.1008/(0.1008 + 0.0021) \approx 0.980$. Classified as spam.

Part C

19. (a) $P(M_1) = 0.6$, $P(M_2) = 0.4$, $P(D \mid M_1) = 0.03$, $P(D \mid M_2) = 0.05$ (b) $P(D) = 0.03(0.6) + 0.05(0.4) = 0.018 + 0.020 = 0.038$
20. $P(\text{urgent}) = 0.3(0.4) + 0.1(0.35) + 0.6(0.25) = 0.12 + 0.035 + 0.15 = 0.305$
21. $P(+) = 0.95(0.02) + 0.10(0.98) = 0.019 + 0.098 = 0.117$
22. $P(-) = 1 - 0.117 = 0.883$; directly: $P(- \mid D) \cdot P(D) + P(- \mid D^c) \cdot P(D^c) = 0.05(0.02) + 0.90(0.98) = 0.001 + 0.882 = 0.883$. ✓
23. $P(\text{cancelled}) = 0.05(0.5) + 0.30(0.3) + 0.90(0.2) = 0.025 + 0.09 + 0.18 = 0.295$
24. B_i are the clusters $Z = k$; $P(A \mid B_i)$ is $P(\mathbf{x} \mid Z = k)$; $P(B_i)$ is the mixture weight $P(Z = k)$.
25. (a) $P(T) = 0.10(0.5) + 0.20(0.3) + 0.05(0.2) = 0.05 + 0.06 + 0.01 = 0.12$ (b) Apply Bayes: $P(B \mid T) = 0.06/0.12 = 0.5$ — Road B most likely.
26. Partitions have $P(A \mid B_1) = 1/2$, $P(A \mid B_2) = 1/2$, $P(A \mid B_3) = 1/2$; $P(A) = 1/2 \cdot 1/3 + 1/2 \cdot 1/3 + 1/2 \cdot 1/3 = 1/2$. ✓

Part D

27. $P(M_1 \mid D) = \frac{0.018}{0.038} \approx 0.474$ (about 47%)
28. $P(D \mid +) = \frac{0.019}{0.117} \approx 0.162$ — only 16%! (a) Most positives are false positives when prevalence is low. (b) With 20% prevalence: $P(+) = 0.95(0.2) + 0.10(0.8) = 0.19 + 0.08 = 0.27$; $P(D \mid +) = 0.19/0.27 \approx 0.704$.
29. $P(\text{spam} \mid \text{urgent}) = \frac{0.6(0.25)}{0.305} = \frac{0.15}{0.305} \approx 0.492$; $P(\text{work} \mid \text{urgent}) = \frac{0.3(0.4)}{0.305} \approx 0.393$.
30. $P(+) = 0.9(0.05) + 0.01(0.95) = 0.045 + 0.0095 = 0.0545$; $P(O \mid +) = 0.045/0.0545 \approx 0.826$.
31. New prior $= 0.85$. $P(+\text{winner} \mid \text{spam}) = 0.9(0.85) = 0.765$; $P(+\text{winner} \mid \neg) = 0.1(0.15) = 0.015$; updated: $0.765/(0.765 + 0.015) \approx 0.981$.
32. $P(\text{Roman} \mid \text{test}) = \frac{0.7(0.6)}{0.7(0.6)+0.2(0.4)} = \frac{0.42}{0.42+0.08} = \frac{0.42}{0.50} = 0.84$. Medieval: 0.16.
33. (a) 1/3 (b) With the standard host rule: $P(C_1 \mid H_3) = 1/3$, $P(C_2 \mid H_3) = 2/3$. (c) Yes, switch — doubles the probability of winning.

34. When comparing classes, dividing by the same denominator $P(B)$ doesn't change the argmax — so classification only requires the numerator.

35. $P(\text{class})$ = prior; $P(w_i \mid \text{class})$ = likelihood per word; product uses Bayes via independence assumption; "naïve" because words are assumed conditionally independent, which is unrealistic.

36. Spam log-score: $\ln(0.1008) \approx -2.295$; not-spam: $\ln(0.0021) \approx -6.170$. Spam wins — same classification.

Part E

37. Verify: the prior $P(\text{rain})$; whether dark clouds are actually a reliable signal; whether the model was trained on similar conditions.

38. Example: a disease screening with 0.1% prevalence and a test with 99% sensitivity/specificity — the majority of positive results are still false positives.

39. (a) $\pi_j^{(t+1)} = \sum_i \pi_i^{(t)} P_{ij}$, which is exactly $P(\text{state } j \text{ at } t{+}1) = \sum_i P(\text{came from } i) \cdot P(j \mid i)$. (b) The stationary distribution is the one where these long-run proportions remain constant.

40. (Answers will vary) Bayes' theorem updates a prior belief using new evidence to produce a posterior belief. It says: the probability of a hypothesis given evidence equals the likelihood of the evidence under the hypothesis, times the prior, divided by the overall probability of the evidence. It is the mathematical foundation for how ML models learn from data.